PHY 101

REVISION QUESTIONS AND ANSWERS

VOTE

IFEKITAN 2019

As PUBLIC RELATIONS OFFICER

FEDERAL UNIVRSITY OF TECHNOLOGY
AKURE STUDENT UNION

ALUKORO TO KAJUE || THE CAPABLE
TOWN CRIER

Q1 in a perfectly inelastic collision

a) heat is gained

b) energy is lost

c) energy is gained

d) power is lost

Q2 in an inelastic collision which of the following is not true?

a) kinetic energy is not conserved

b) kinetic energy is conserved

c) potential energy is conserved

d) potential energy is not conserved

Q3 The collision between two or more objects during which no energy is lost is called

a) conserved collision

b) material collision

c) inelastic collision

d) elastic collision

Q4 The sudden impact felt between two or more objects is called

a) explosion

b) conservation

c) collision

d) None of these

Q5 Linear momentum is always conserved when the rate of change of momentum is

a) zero

b) one

c) linear

d) circular

Q6 Work is a product of

a) force x distance

b) force x time

c) force x momentum

d) force x velocity

Q7 The path followed by the projectile is known as

a) curve

b) range

c) flight time

d) Trajectory

Q8 The range of a projectile launched from the ground is the

a) vertical distance

b) perpendicular distance

c) horizontal distance

d) None of these

Q9 In a projectile motion, at maximum height the verticle component of velocity is

a) greatest

b) zero

c) downwards

d) upwards

Q10 Projectile motion is an example of motion in how many dimensions?

a) one

b) two

c) three

d) four

Q11 The transfer of heat through solids is known as

a) **Conduction**
b) radiation
c) convection
d) Thermodition

Q12 The constant -volume gas thermometer is normally refered to as the

a) mercury thermometer
b) liquid thermometer
c) **standard thermometer**
d) gas thermometer

Q13 Force is equal to the product of mass and

a) **acceleration**
b) velocity
c) time
d) distance

Q14 Which of the following is the SI Unit of momentum

a) N/s
b) kgm
c) N
d) **kg/m**

Q15 The product of force and time is equal to

a) momentum
b) watts
c) acceleration
d) **impulse**

Q16 A symmetrical body rotating about an axis with one point fixed at the centre of gravity is called ----------

a) Top

b) gyroscope

c) rotor

d) None of these

Q17 A body starts from rest and accelerates uniformly at a rate of 5m/s2. calculate its velocity after 90m

a) 18m/s

b) 30m/s

c) 25m/s

d) 13m/s

Q18 A ball is thrown vertically upwards from the ground with an initial velocity of 50m/s. The total time spent by the ball in the air will be -----------

a) 10sec

b) 15sec

c) 20sec

d) 25sec

Q19 A man will exert the greatest pressure on a bench when he ----------

a) lies flat on his back

b) lies flat on his belly

c) stands on the toes of one foot

d) stand on both feets

Q20 The slope of a straight line displacement time graph gives the --------------

a) distance travelled

b) uniform velocity

c) acceleration

d) uniform speed

Q21 The force of ---------------allow cars to move on the roads without skidding

a) engine

b) centripetal

c) centrifugal

d) friction

Q22 Acceleration due to gravity is ------------- at the poles than at the equator

a) minimal

b) tighter

c) greater

d) lost

Q23 The --------------- velocity is the minimum velocity needed by an object to be project into space from the surface of the earth.

a) escape

b) vertical

c) initial

d) final

Q24 Acceleration due to gravity increases with -------------

a) magnitude

b) altitude

c) latitude

d) longitude

Q25 Below the earth's surface, the accelaration due to gravity decreases linearly with

a) density

b) depth

c) height

d) None of these

Q26 which of the following is not a fundamental force

a) gravitational

b) electroweak

c) strong

d) minute

Q27 The acceleration due to gravity vary from location to location because of the effect of earth's

a) rotation

b) radius

c) weight

d) mass

Q28 The √2gr is known as the

a) uniform velocity

b) escape velocity

c) fixed velocity

d) constant velocity

Q29 The height of the parking orbit above the surface of the is given by

a) h=R+r

b) h=R-V

c) h=R-r

d) h=r-R

Q30 The internal energy U of a system depends on the

a) power of the system

b) state of the system

c) energy equation

d) internal resistance

Q31 The velocity of orbiting satellite is given by

a) √2gr

b) √GM

c) r/t

d) $\sqrt{GM/r}$

Q32 The gravitational force on a sattelite produces the centripetal acceleration that keeps the satellite in

a) orbit

b) constant position

c) radial path

d) freefall

Q33 The path describe by a satellite round the sun is a

a) conical section

b) spherical

c) full circle

d) None of these

Q34 When there is no reaction force to an objects weight, the object becomes

a) strong

b) weightless

c) gravitating

d) freefall

Q35 The path of a satellite whose period of revolution is equal to the period of rotation of the earth about its axis is known as

a) rotating orbit

b) stationary orbit

c) parking orbit

d) space orbit

Q36 The process in which energy is transfered by means of electromagnetic waves is known as

a) Conduction

b) convection

c) **radiation**

d) resistation

Q37 1kg mass placed on the moon will weigh

a) 9.8N

b) 9.7N

c) 1.67kg

d) **1.67N**

Q38 Which of the quantities vary from point to point on the surface of the earth

gravitational constant

a) mass

b) moon

c) **weight**

Q39 Energy expended per time is called

a) **power**

b) work

c) joule

d) kinetic

Q40 The quantity of matter contained in a body is called

a) **weight**

b) mass

c) matter

d) gravity

Q41 Newton's second law states that the rate of change of momentum is directly proportional to the

a) pressure

b) **force**

c) energy

d) power

Q42 Stable equilibrium happens when the forces in the displaced position act such that they return the body to its

a) original position

b) unstable position

c) final position

d) neutral position

Q43 In a circular motion which of the following quantities is constant

a) velocity

b) time

c) mass

d) accelaration

Q44 When a body accelerates in motion it means that its

a) path is linear

b) velocity is uniform

c) velocity changes continously

d) None of these

Q45 The velocity of a particle at some point of its path is called

a) uniform velocity

b) instantaneous velocity

c) regular velocity

d) All of these

Q46 Velocity is defined as

a) distance/time

b) speed/time

c) displacement/time

d) variation/time

Q47 A car that takes 2hrs to travel from kaduna to kagoro along a winding road over a distance of 200km is said to have an average speed of

a) 100km/hr

b) 10km/hr

c) 1000km/hr

d) 10,000km/hr

Q48 If you run on a winding path from point A to point B and travel a distance of 240m in 20sec then your average speed is

a) 6m/s

b) 12m/s

c) 16m/s

d) 20m/s

Q49 Which of the following is a vector quantity?

a) distance

b) speed

c) temperature

d) displacement

Q50 The result of the cross product of two vectors is a

a) scalar

b) zero quantity

c) determinant

d) vector

Q51 The scalar product of any two vectors at right angles to each other is always

a) 1

b) 90

c) zero

d) None of these

Q52 The degree of hotness or coldness of a body is called

a) pressure

b) heat

c) temperature

d) kelvin

Q53 if Vector P = 5i -2j and Q=3i+3j and R=4i-j . What is P+Q+R?

a) 6i

b) 12i

c) 7i

d) 14i

Q54 A Null vector is a vector whose magnitude is

a) zero

b) very large

c) continous

d) available

Q55 If a force of 40N acting in the direction due East and a force of 30N is acting in the direction due North. Then the magnitude of the resultant forces will be

a) 40N

b) 30N

c) 20N

d) 50N

Q56 A Scalar quantity has only

a) direction

b) velocity

c) magnitude

d) time

Q57 A Vector quantity has both direction and

a) unit

b) magnitude

c) speed

d) quantity

Q58 the change in the thermo-electric e.m.f per degree Celsius in temperature between the hot and cold junctions is known as

a) clinical power

b) thermo-electric power

c) electric

d) power

Q59 The dimension of coefficient of linear expansion is

a) K

b) degrees

c) Lxdegrees

d) L

Q60 The thermometer that is constructed by using the Seebeck effect is known as

a) Thermo-electric

b) clinical

c) mercury

d) None of these

Q61 Which of the following gives the dimension of velocity?

a) TxLxL

b) LxT

c) L/T

d) L/M

Q62 Which of the following will give the dimension of Area

a) LxL

b) LxLxL

c) MxM

d) MxMxM

Q63 The SI Unit of time is

a) minute

b) seconds

c) day

d) year

Q64 Which of the following quantities have a derived unit?

a) length

b) time

c) area

d) mass

Q65 The rope shown in the figure attached hereto is wound around a cylinder of of mass 4.0 kg, radius 10cm and

I=0020kgm2 Is the moment of inertia about an axis along the cylinder axis. If the cylinder rolls without slipping, what the linear acceleration of its centre of mass?

A. 6.7m/s2

B. 3.3m/s2

C. 11.2m/s2

D. 1.5m/s2

Q66 The frictional force between the block and the table is 20 N. If the moment of inertia of the wheel is

4.0kgm2, find how long it will take the block to drop

60cm after the system is released.

A. 7.37 s

B.1.34 s

C. 3.71 s

D. 4.22 s

Q67. The mass shown in the figure attached starts from rest and falls a distance h. Find the angular speed of the wheel in terms of the radius of the wheel and I, m and h.

A. ω=(2mgI/mb2−h)1/2

B. ω=(2mgI/mb2+h)1/2

C. ω=(2mgh/mb2−I)1/2

D. ω=(2mgh/mb2+I)1/2

Q68. Find the equation for the simple harmonic motion shown

A. y=0.08sin(0.40πt−0.40π)

y=0.06sin(0.80πt+0.40π)

C. y=0.08sin(0.40πt+0.40π)

D. y=0.06sin(0.80πt−0.40π)

Q69 For the simple harmonic shown, evaluate y at t=1.35s

A. 0.034 m

B. 0.023 m

C. 0.047 m

D. 0.012 m

Q70 When a 30-g mass is hung from the end of a spring, the spring stretches 8.0 cm. The same spring with a mass of 200 g at at its end is stretched 5.0

cm, released and allowed to oscillate on a frictionless horizontal surface. Find the frequency of the oscillation.

A. 0.54 Hz

B. 0.68 Hz

C. 0.34 Hz

D. 9.5 Hz

Q71 The system shown is an example of the Atwood's machine. What is the acceleration of the masses? Assume the the pulley is frictionless and the rope massless. Take g=9.8m/s2

A. 4.2m/s2

B. 2.5m/s2

C. 9.8m/s2

D. 3.3m/s2

Q72 A boy intends to move an m-kg crate across the floor by applying a constant force P newtons on it.The coefficient of friction between the floor and the crate is μ. Which of these is the best option for his task?

A. Push the crate with P applied horizontally

B. Push the crate with P inclined at an angle above the horizontal

C. **Pull the crate with P inclined at an angle above the horizontal**

D. Pull the crate with P inclined an angle below the horizontal

Q73 A boat propelled so as to travel with a speed of 0.50m/s in still water, moves directly (in a straight line perpendicular to the banks of the river assumed to be straight) across a river that is 60m wide. The river flows with a speed of 0.30m/s. How long in seconds does it take the boat to cross the river?

A. 150

B. 120

C. 300

D. 200

Q74 Which of the follwing statements is correcct?

A. An object can have a constant velocity even though its speed is changing

B. An object can have a constant speed even though its velocty is changing

C. An object can have zero acceleration and eventually reverses its direction

D. An object can have constant velocity even though its acceleration is not zero

Q75 Which of these is NOT a statement of Newtonâ€™s law of universal gravitation?

A. gravitational force between two particles is attractive as well as repulsive

B. gravitational force acts along the line joining the two particles

C. gravitational force is directly proportional to the product of the masses of the particles

D. gravitational force is inversely proportional to the square of the distance of the particles apart

Q76 How large an average force is required to stop a 1400-kg car in 5.0 s if the crate’s initial speed is 25 m/s?

A. 2000 N

B. 3500 N

C. 9000 N

D. 7 000N

Q77 A 10-g bullet of unknown speed is shot horizontally into a 2-kg block of wood suspended from the ceiling by a cord. The bullet hits the block and becomes lodged in it. After the collision, the block and the bullet swing to a height 30cm above the original position. What was the speed of the bullet?

(This device is called the ballistic pendulum). Take g=9.8ms−2

A. 487 m/s

B. 640 m/s

C. 354 m/s

D. 700 m/s

Q78 A 40-g ball travelling to the right at 30 cm/s collides head on with an 80-g ball that is at rest. If the collision is perfectly elastic, find the velocity of each ball after collision

A. the first ball is going to the right at 10m/s while the other is going to the left at 20m/s

B. the first ball is going to the left at 10m/s while the other is going to the right at 20m/s

C. the first ball is going to the left at 20 m/s while the other is going to the right at 10 m/s

D. the first ball is going to the right at 10 m/s while the other is going to the left at 10 m/s

Q79 A gun of mass M is used to fire a bullet of mass m. The exit velocity of the bullet is v. Find the recoil velocity of the gun

A. Mv/m

B. mv/M

C. −Mv/m

D. −mv/M

Q80 A 30,000-kg truck travelling at 10.0m/s collides with a 1700-kg car travelling at 25m/s in the opposite direction. If they stick together after the collision, how fast and in what direction will they be moving?

A. 8.1 m/s in the direction of the truck’s motion

B. 12.3 m/s in the direction of the car’s motion

C. 24.2 m/s in the direction of the car's motion

D. 17.6 m/s in the direction of the truck's motion

Q81 Sand drops at the rate of 2000 kg/min. from the bottom of a hopper onto a belt conveyor moving horizontally at 250 m/min. Determine the force needed to drive the conveyor, neglecting friction.

A. 500 N

B. 800 N

C. 139 N

D. 152 N

Q82 The exhaust gas of a rocket is expelled at the rate of 1300 kg/s, at the velocity of 50 000 m/s. Find the thrust on the rocket in newtons

A. 6.5×107

B. 3.5×107

C. 7.6×107

D. 5.7×107

Q83 A force of $2\vec{i}$ $+7\vec{j}$ N acts on a body of mass 5kg for 10 seconds. The body was initially moving with constant velocity of $\vec{i}$ $2\vec{j}$ m/s. Find the final velocity of the body in m/s, in vector form.

A. $5\vec{i}$ $+12\vec{j}$

B. $12\vec{i}$ $5\vec{j}$

C. $10\vec{i}$ $7\vec{j}$

D. $7\vec{i}$ $+10\vec{j}$

Q84 Two trolleys X and Y with momenta 20 Ns and 12 Ns respectively travel along a straight line in opposite directions before collision. After collision the directions of motion of both trolleys are reversed and the magnitude of the momentum of X is 2 Ns. What is the magnitude of the corresponding momentum of Y?

A. 6 Ns

B. 8 Ns

C. 10 Ns

D. 30 Ns

INFORMATION IS LIGHT, I CHOOSE TO CARRY IT

IFEKITAN'19 AS FUTASU PRO

PHY101 TEST ELUCIDATION

✍ Written by Ajele Ifeoluwa Emmanuel

This course is quite easy, but people have the wrong mindset about it. I had B in my physics 101 even when I wasn't so good with calculation.

In your mind ask how?

Here's the gimmick. Physics isn't all about calculation alone. Redirect your focus. Mostly normal theoretical objective questions of words are more than calculation based questions in your test and examination. So be properly guided, it's only if you are unlucky will calculation be more than normal wordy questions.

• Read the handout first and understand it, physics is something practical, relate it to something, understand the note. Even when you get to a calculation part that you don't seem to understand, star it and move on to reading. Later when you come for tutorials which I will organise this week and probably when you see someone to explain to you, the focus will just be on the calculation aspect.

So get it right from now, physics isn't just about calculation.

YOUR TEST WILL MOST LIKELY COVER THE FOLLOWING TOPICS

- **Dimension Analysis**
- **Vector Analysis**
- **Kinematics**
- **Projectile Motion**
- **Circular Motion**
- **Laws of Mechanic**

The questions will be of six sections, five questions per section.

Let me sound a note of warning here.

Physics department is very particular about dressing and looks. The lecturers are always on ground during the test. So guys, please no dreadlocks, ensure your hair is well combed and ladies endeavor not to put on armless tops not crazy Jean to avoid undue embarrassment.

A word is enough for the wise. Best of luck.

Don't forget to Vote IFEKITAN AS THE FUTASU PRO in the forthcoming election.

God bless you

Information is light and I choose to carry it